

Product Reference Manual (V1.0)

Description

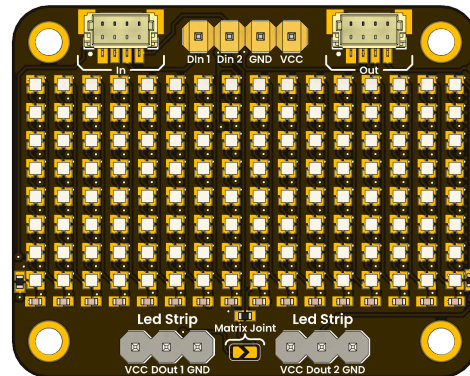
The **DevLab 8×16 RGB LED Matrix** is a compact and versatile display module built around **128 individually addressable WS2812B LEDs**, arranged in an **8-row by 16-column** grid. Each LED contains an integrated driver circuit, enabling full 24-bit color control (8 bits per RGB channel) through a single-wire digital interface.

This module is designed for seamless integration in embedded and DIY lighting projects, offering rich color rendering and dynamic visual effects with minimal wiring complexity. The onboard **dual-channel input/output** configuration (DIN1/DOUT1 and DIN2/DOUT2) allows cascading multiple panels or LED strips in parallel or series. Standard **JST-SM and pin-header connectors** simplify wiring and chaining, while the **Matrix Joint interface** supports modular panel extensions for creating larger displays.

Its **standard I/O voltage (5 V logic)** and **wide compatibility** make it ideal for use with microcontrollers such as ESP32, Arduino, Raspberry Pi, and STM32.

DevLab format compatibility.

Simplicity and compatibility are the core principles of the **DevLab form factor**. This standard defines a **compact and communication-optimized board layout**, ensuring straightforward connection and interoperability among DevLab module



Key Features

- **Key Features**
- 8×16 grid (128 individually addressable RGB LEDs)
- Integrated WS2812B smart LEDs with built-in drivers
- Full 24-bit color control (8-bit per RGB channel)
- Single-wire serial data input (800 kHz protocol)
- Dual data channels (DIN1/DOUT1 and DIN2/DOUT2) for flexible control
- JST-4 pin connectors for easy input/output chaining
- 4-pin header for external wiring (DIN1, DIN2, GND, VCC)
- “Matrix Joint” expansion header for linking multiple panels
- Operates from 4.5 V – 5.5 V (typical 5 V supply)
- High brightness and wide viewing angle (>120°)
- Compatible with Arduino, ESP32, STM32, and Raspberry Pi
- Individual pixel addressing for text, graphics, and animations
- Gold-plated PCB with black solder mask for durability
- Compact form factor with M3 mounting holes
- Typical current: ~20 mA per LED (average brightness)

- Up to 60 mA per LED at full white brightness

Software Support

- **Arduino IDE**
- **Micropython**

Applications

- Animated logos, symbols, and text displays
- Visual indicators for IoT or robotics projects
- Music visualizers and sound-reactive lighting
- Wearable and artistic light installations
- Prototyping of UX/UI lighting systems
- Smart decoration and ambient illumination
- DIY electronic projects and educational kits

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1 The Board

1.1 Accessories

- 1×6-pin 2.54 mm male header

2 Ratings

2.1 Recommended Operating Conditions

Parameter	Description	Min	Typ	Max	Unit
Vdd	Input voltage to power on the module	3.3	-	5.5	V
Idd*	Supply current (All LEDs at maximum power at 5V)	-	-	2.56	A
	All LEDs at maximum power at 3.3V	-	-	1.68	A
Vdi	Logic input Voltage	-0.4	-	VDD+0.4	V
Vih	Input high threshold voltage	2.8	-	-	V
Vil	Input Low threshold voltage	-	-	1.6	V
Loh	DOUT output current	-	-35	-	mA
Lol	DOUT current injection	-	35	-	mA
Fpwm	Output PWM frequency	-	1	-	KHz
Iv	Light Intensity Red	-	200	-	mcd
	Light Intensity Green	-	800	-	mcd

Parameter	Description	Min	Typ	Max	Unit
	Light Intensity Blue	-	300	-	mcd
λD	Main Wavelength Red	620	-	630	nm
	Main Wavelength Green	520	-	530	nm
	Main Wavelength Blue	465	-	475	nm
$2\theta 1/2$	Half light angle	-	120	-	Deg

***Note:** When powering the LEDs at 5V, their temperature may rise above 100°C at higher power levels. It is strongly recommended to implement optimal thermal management or to limit the use of LEDs at high brightness. Also, be careful when powering via the JST connector; it is recommended to limit the operating current to less than 3A.

3 Functional Overview

The **DevLab WS28B20-1010 RGB LED Matrix Module** is a digitally addressable display subsystem designed for embedded visual output, lighting control, and human-machine interface applications. The module integrates **128 WS28B20 RGB LEDs** arranged in a fixed **8 × 16 matrix topology**, providing deterministic pixel addressing and uniform spatial mapping.

Each WS28B20 device contains an internal control IC and constant-current RGB drivers. This architecture eliminates the need for external LED drivers or multiplexing circuitry, allowing the matrix to be controlled through a **single-wire, time-encoded digital protocol**. Pixel data is latched internally and refreshed continuously, ensuring stable color reproduction without MCU intervention once transmission is complete.

3.1 Data Flow Architecture

Pixel data is transmitted serially and propagates LED-to-LED following the physical routing defined on the PCB. The first LED in the chain (**LED #1**) captures the first 24-bit RGB frame and forwards remaining data downstream until all LEDs are updated, terminating at **LED #128**.

The PCB silkscreen clearly marks:

- Data direction
- First and last LED positions
- Matrix segmentation boundaries

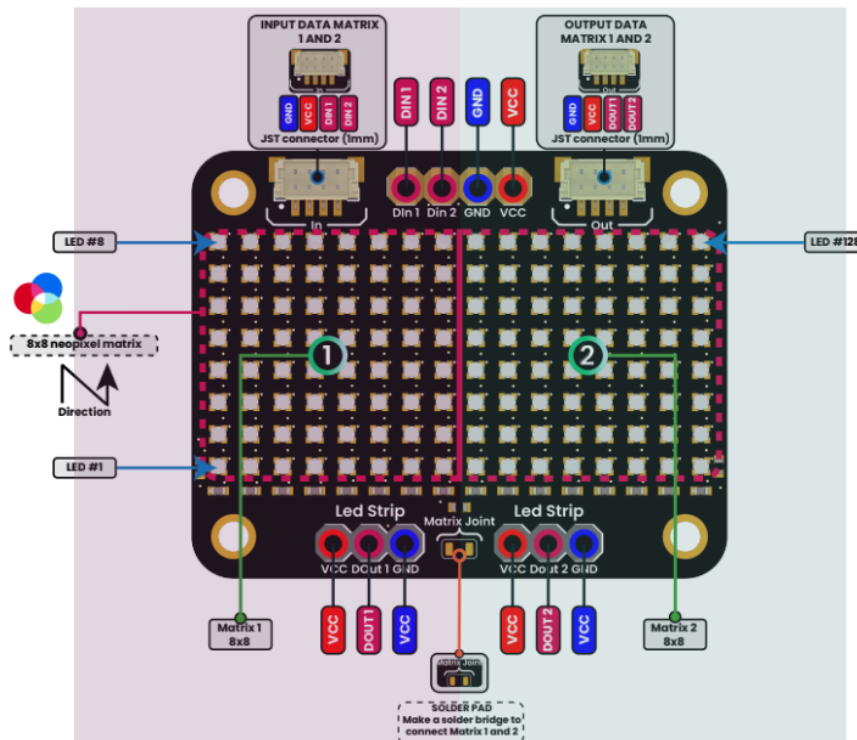
This deterministic data flow enables:

- Predictable animation timing
- Linear memory mapping
- Efficient DMA-based or interrupt-free data streaming on capable MCUs

3.2 Dual-Matrix Operation Concept

Internally, the board is partitioned into **two independent 8 × 8 sub-matrices**:

- **Matrix 1:** LEDs #1–#64
- **Matrix 2:** LEDs #65–#128



Each sub-matrix has its own dedicated:

- Data input (DIN1 / DIN2)
- Data output (DOUT1 / DOUT2)

This architecture allows two distinct operational modes:

Independent Matrix Mode

- Matrix 1 and Matrix 2 are driven by separate data signals
- Enables parallel animations, mirrored effects, or dual-content rendering
- Ideal for stereo visualizers, segmented indicators, or status panels

Unified Matrix Mode

- A dedicated **Matrix Joint solder pad** electrically links Matrix 1 and Matrix 2
- Forms a single continuous **8 × 16 LED chain**
- Only one data input is required
- Simplifies software addressing and animation logic

3.3 Expansion and Cascading

The module includes **data output paths** that preserve signal timing and protocol integrity, allowing additional LED matrices or LED strips to be connected in series.

Supported expansion topologies:

- Matrix-to-matrix cascading

- Matrix-to-strip chaining
- Mixed matrix + strip configurations

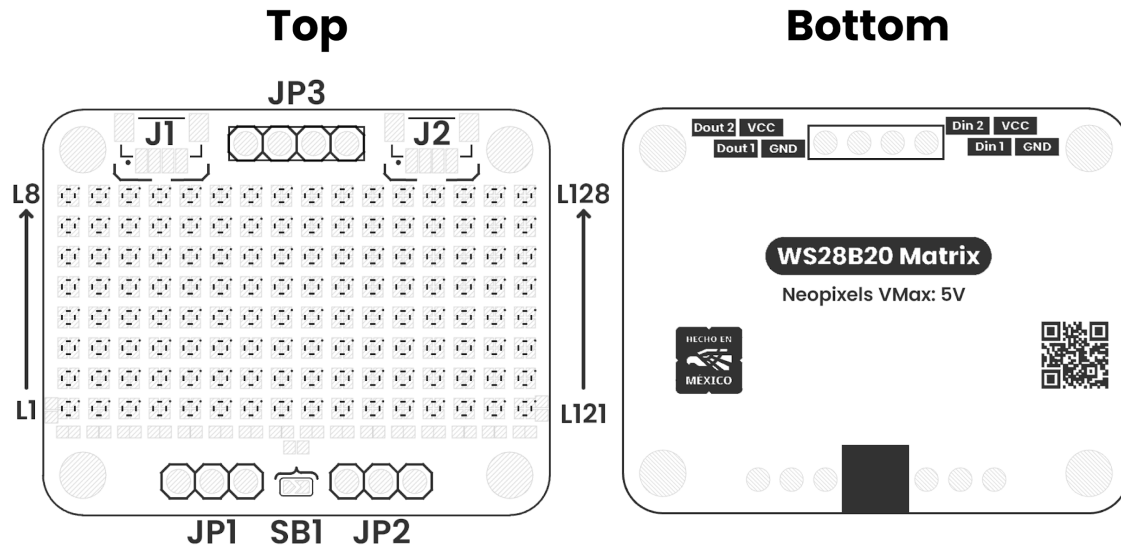
Power rails are routed alongside data outputs to simplify wiring, though additional power injection is recommended for large cascaded systems.

3.4 Electrical and Signal Behavior

- Data signal is **non-return-to-zero (NRZ)** encoded
- Timing-sensitive; requires precise pulse widths
- Compatible with bit-banging, hardware peripherals, or dedicated LED engines
- Data line is **unidirectional**
- LEDs retain last state when no new data is received

The matrix does not require clock synchronization, external latches, or refresh timers.

3.2 Board Topology



Views of Board Topology

Views of DevLab Topology

Table 3.2.1 - Components Overview

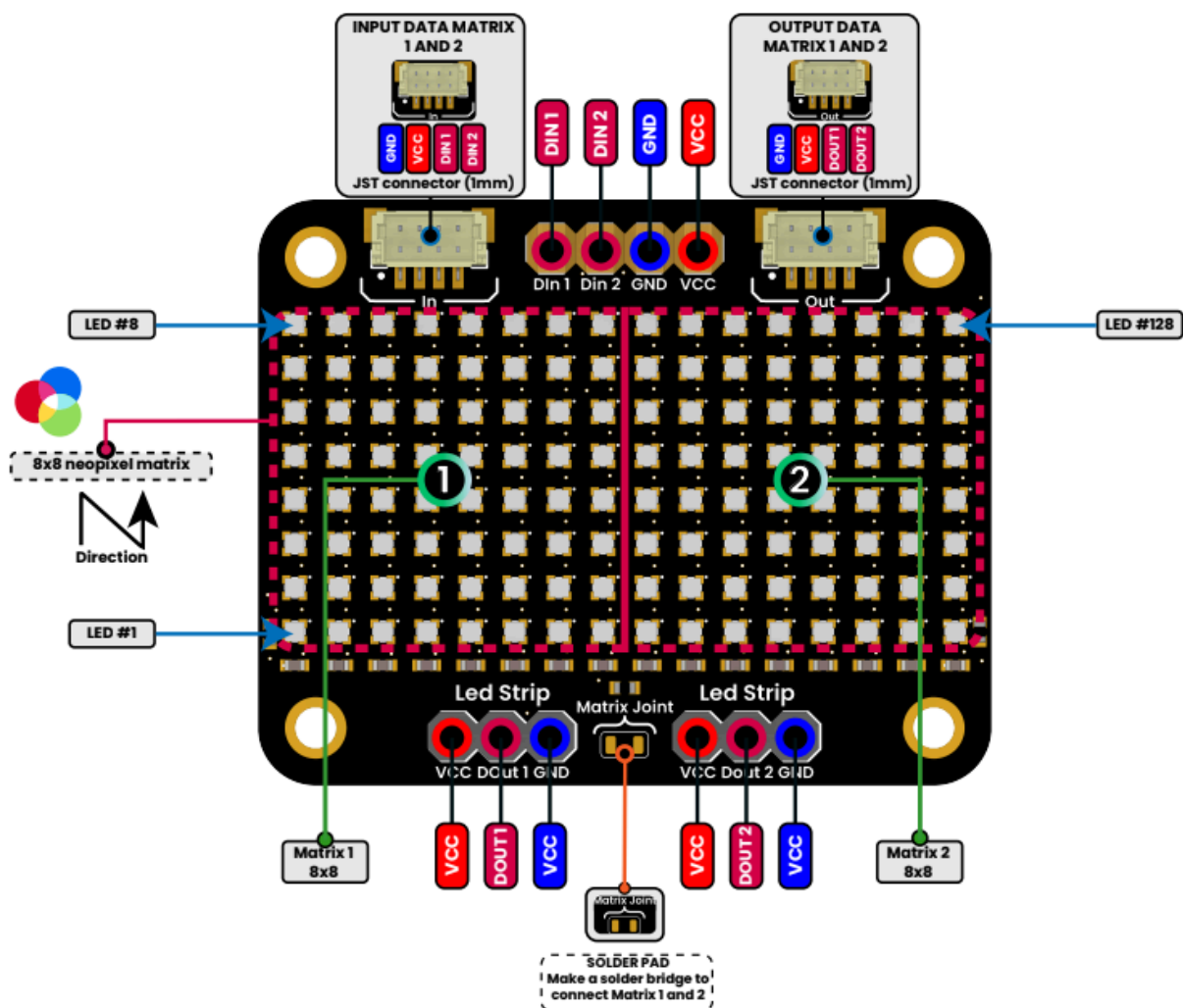
Ref.	Description
JP1	2.54 mm Pin Header
JP2	2.54 mm Pin Header
JP3	2.54 mm Pin Header
J1	JST 1 mm Pitch Connector for Data Input
J2	JST 1 mm Pitch Connector for Data Output
L1-L64	WS28B20 Neopixel for Matrix 1
L65-L128	WS28B20 Neopixel for Matrix 2
SB1	Solder Bridge to Join Both Matrices

4 Connectors & Pinouts

4.1 General Pinout

PINOUT

DevLab: WS28B20-1010 RGB LED Matrix Module



DevLab General Pinout

4.2 Pinout General Description

Pin Label	Function / Notes
VCC	5 V power supply input for the LED matrix. This rail powers all WS28B20 LEDs. An external power supply is recommended when operating at medium or high brightness levels.
GND	Ground reference. Must be electrically common with the controlling microcontroller ground to ensure correct data signaling.
DIN1	Digital data input for Matrix 1 (left 8×8 section). Receives serialized RGB data following the WS28B20/NeoPixel protocol.
DIN2	Digital data input for Matrix 2 (right 8×8 section). Allows independent control when the matrix is operated in split mode.
DOUT1	Digital data output from Matrix 1 . Used to cascade additional LED matrices or external NeoPixel-compatible LED strips.
DOUT2	Digital data output from Matrix 2 . Enables parallel expansion or independent downstream LED chains.
Matrix Joint	Solder-bridge pad used to electrically connect Matrix 1 and Matrix 2. When bridged, the module operates as a single 8×16 matrix driven from a single data input.
LED Strip VCC	5 V power output provided at the LED strip pads for powering external NeoPixel-compatible devices.
LED Strip GND	Ground output for external LED strips or cascaded modules.
LED Strip DOUT	Data output signal for driving external LED strips or chained matrices.

Pinout Notes

- *Logic Level: WS28B20 devices operate natively at 5 V logic; most 3.3 V MCUs are compatible in short wiring scenarios.*
- *Power Considerations: At full white, each LED can draw up to ~60 mA. Proper power sizing and injection are required for stable operation.*
- *Data Direction: Data flow direction is fixed and indicated on the PCB silkscreen (LED #1 → LED #128).*
- *Interface Redundancy: The same signals are available via pin headers, JST connectors, and solder pads to accommodate different integration methods.*

5 Board Operation

The DevLab WS28B20-1010 module is engineered for **plug-and-play integration** into embedded systems. From a system perspective, the board behaves as a high-current peripheral that requires careful power management but minimal control signaling.

5.1 Electrical Operation Principles

Power Distribution

- Nominal supply voltage: **5 V DC**

Although typical applications operate far below maximum current, system designers must account for peak load conditions during brightness transitions or startup animations.

Recommended practices:

- Use an external 5 V supply for medium to high brightness
- Ensure common ground between MCU and LED matrix
- Inject power close to the matrix when cascading modules
- Avoid powering the matrix directly from MCU 5 V pins

Signal Integrity

- Data input should be a clean digital signal
- Short data lines are preferred
- Series resistors (220–470 Ω) may be used for signal conditioning if required
- Level shifting may be necessary when driving from 3.3 V-only MCUs in noisy environments

5.2 Board Configuration Options

Configuration	Description
Single Input	One DIN controls entire 8×16 matrix
Dual Input	Independent control of two 8×8 matrices
Cascaded Output	Drive additional matrices or LED strips
Split Operation	Matrix used as two logical displays

Matrix configuration is hardware-defined and does not require firmware changes beyond LED count and indexing.

5.1 Getting Started with Arduino IDE

This section describes the **baseline operational workflow** using the Arduino ecosystem. The same principles apply to other platforms.

5.1.1 Development Environment

Software Requirements

- Arduino IDE (latest stable release)
- NeoPixel-compatible LED control library
- USB driver for selected MCU

Hardware Requirements

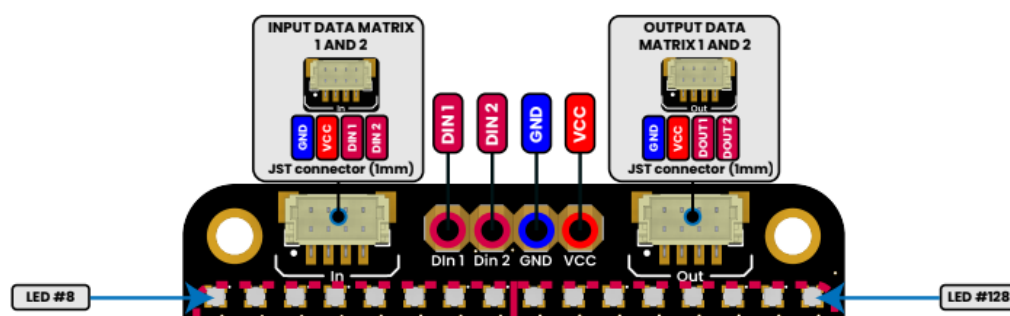
- Arduino-compatible microcontroller
- DevLab WS28B20-1010 RGB LED Matrix Module
- External 5 V power supply (recommended)
- Common ground connection

5.1.2 Initial Wiring Configuration

Matrix Signal	Arduino Connection	Purpose
VCC	5 V (external)	LED power
GND	GND	Reference ground
DIN1	Digital GPIO	Data input

If using unified mode, only **DIN1** is required.

For independent mode, **DIN1** and **DIN2** may be connected to separate GPIO pins.



5.1.3 Software Initialization Concept

From a firmware perspective, the WS28B20-1010 RGB LED Matrix Module is treated as a linear array of addressable RGB pixels, regardless of its physical 2D layout. Each LED corresponds to a sequential index in memory, starting from the first LED in the data chain.

The logical two-dimensional behavior (rows and columns) is implemented at the application level through software mapping. This abstraction allows the same firmware approach to be used for single matrices, split matrices, or cascaded configurations.

Typical Initialization Steps

1. Define the total LED count
 - 64 LEDs for a single 8 × 8 matrix
 - 128 LEDs for the full 8 × 16 matrix
2. Specify the data pin and LED protocol
 - Select the MCU GPIO connected to DIN
 - Configure LED type (WS28B20 / NeoPixel, GRB order, 800 kHz)
3. Initialize the LED driver library
 - Allocate internal buffers
 - Prepare timing-critical routines
4. Clear the pixel buffer
 - Ensure all LEDs start in a known OFF state
5. Begin rendering frames
 - Write RGB values to the buffer
 - Transmit data to the matrix

Animations, text rendering, and graphical elements are implemented by mapping 2D coordinates (X, Y) to linear LED indices, based on the physical wiring order of the matrix.

5.1.4 Basic Operation Flow

The runtime operation of the LED matrix follows a deterministic, one-way data flow:

1. The MCU prepares an RGB frame buffer in memory
2. The serialized data stream is transmitted via the DIN line
3. Each LED latches its assigned 24-bit RGB value
4. The display updates immediately after data transmission completes
5. The MCU resumes normal operation until the next frame update

No feedback, acknowledgment, or polling from the matrix is required.

The LEDs retain their last state until new data is received or power is removed.

5.1.5 Example Code (Arduino – RGB Color Cycle)

The following example demonstrates a basic RGB color cycle across the entire 8 × 16 WS2812B / WS28B20 matrix.

All LEDs display the same color simultaneously, making this code suitable for:

- Hardware validation
- Power testing
- Brightness adjustment

```

/*
 * Ejemplo simple RGB para Matriz LED WS2812B 8x16
 * UNIT ELECTRONICS - RGB LED Matrix Module
 *
 * Este ejemplo muestra un ciclo continuo de colores RGB en toda la
 * matriz
 */

#include <Adafruit_NeoPixel.h>

// Configuración del hardware
#define PIN        6          // Pin de datos de la matriz
#define NUM_LEDS   128        // Total de LEDs (8x16 = 128)
#define BRILLO     64         // Brillo (0-255)

// Crear objeto NeoPixel
Adafruit_NeoPixel strip(NUM_LEDS, PIN, NEO_GRB + NEO_KHZ800);

void setup() {
  // Inicializar la tira LED
  strip.begin();
  strip.setBrightness(BRILLO);
  strip.clear();
  strip.show();
  // Inicializar comunicación serial (opcional)
  Serial.begin(9600);
  Serial.println("=== Ejemplo RGB Simple ===");
  Serial.println("Matriz 8x16 iniciada");
}

void loop() {
  // Ciclo de colores RGB

```

```
// ROJO
colorCompleto(strip.Color(255, 0, 0));
delay(1000);

// VERDE
colorCompleto(strip.Color(0, 255, 0));
delay(1000);

// AZUL
colorCompleto(strip.Color(0, 0, 255));
delay(1000);

// AMARILLO (Rojo + Verde)
colorCompleto(strip.Color(255, 255, 0));
delay(1000);

// CYAN (Verde + Azul)
colorCompleto(strip.Color(0, 255, 255));
delay(1000);

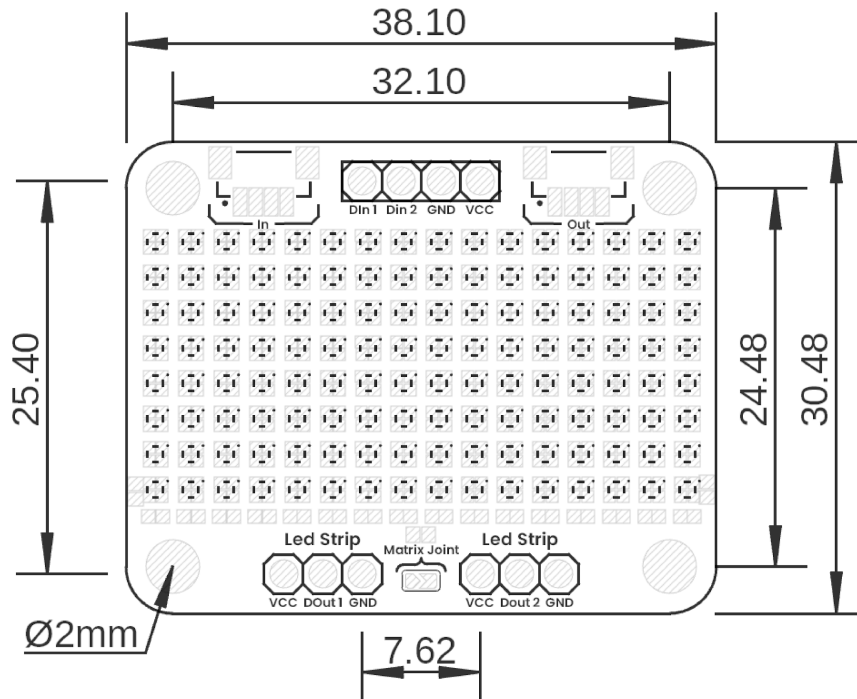
// MAGENTA (Rojo + Azul)
colorCompleto(strip.Color(255, 0, 255));
delay(1000);

// BLANCO (Todos los colores)
colorCompleto(strip.Color(255, 255, 255));
delay(1000);

// APAGADO
colorCompleto(strip.Color(0, 0, 0));
delay(500);
}

// Función para mostrar un color en toda la matriz
void colorCompleto(uint32_t color) {
    for(int i = 0; i < NUM_LEDS; i++) {
        strip.setPixelColor(i, color);
    }
    strip.show();
}
```


6 Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

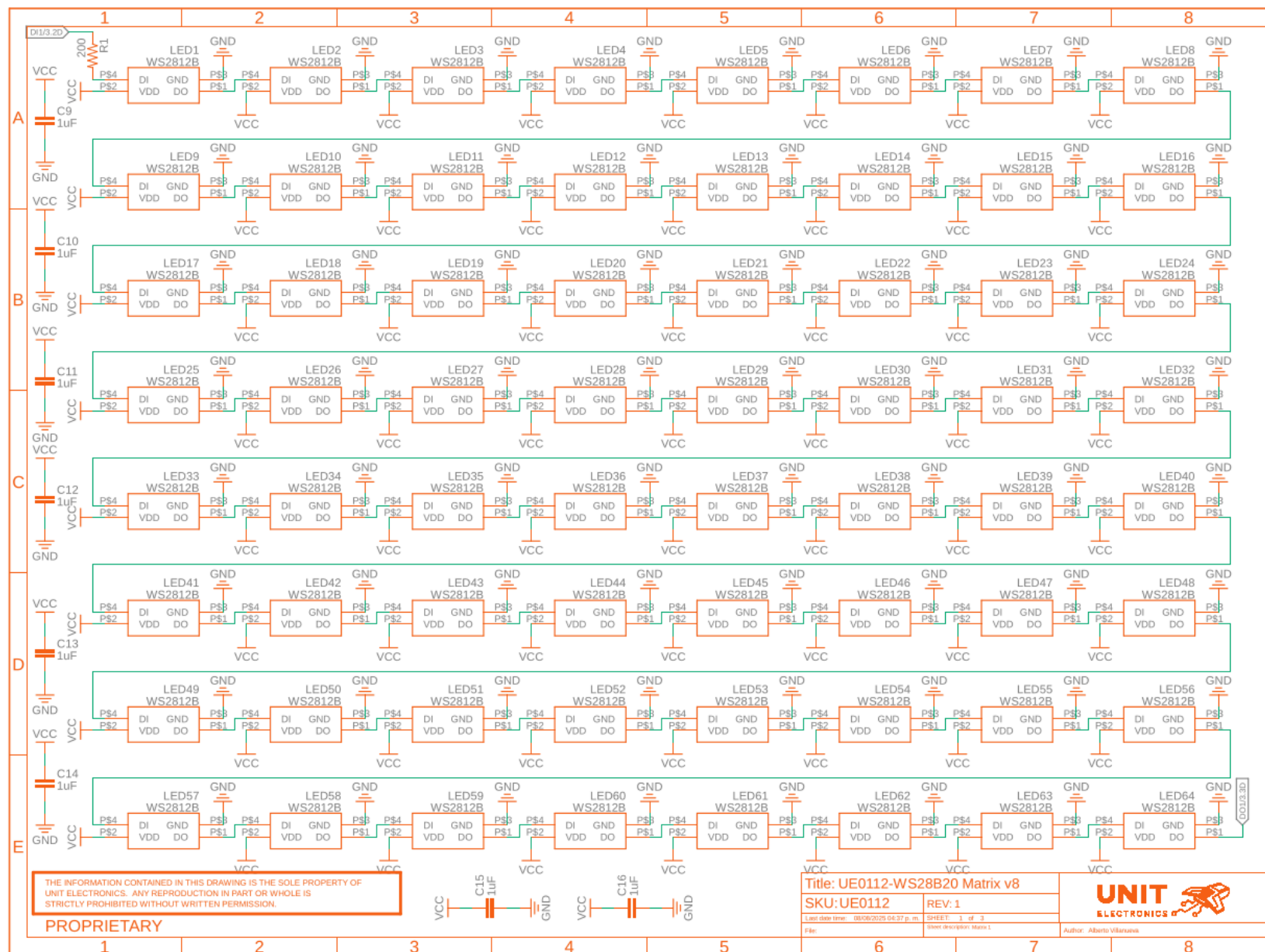
8 Reference Documentation

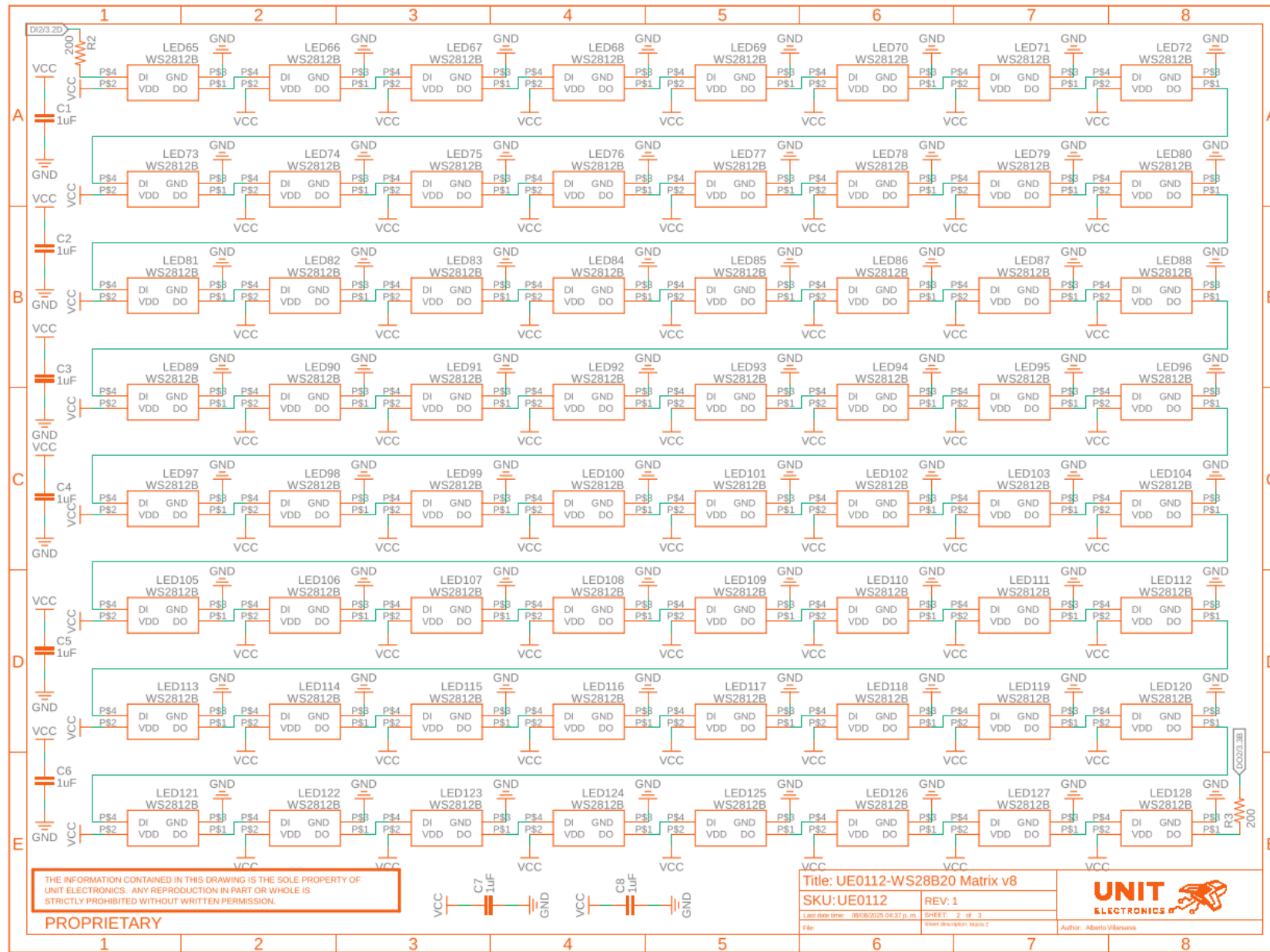
Ref	Link
Documentation	https://github.com/UNIT-Electronics-MX/unit_devlab_ws28b20_1010_rgb_led_matrix_module
Wiki	https://wiki.uelectronics.com/wiki/devlab_ws28b20_1010_rgb_led_matrix_module
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

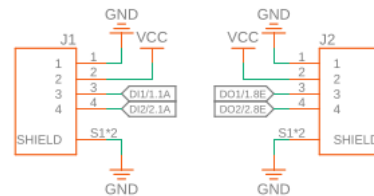
9.1 Schematic

(https://github.com/UNIT-Electronics-MX/unit_devlab_ws28b20_1010_rgb_led_matrix_module/blob/main/hardware/unit_sch_v_1_0_0_ue0112_ws28b20_matrix_rgb_8x16.pdf)





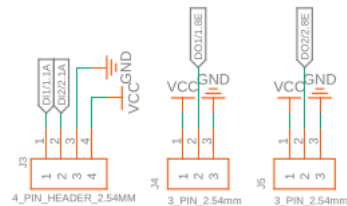
JST Connectors



Solder Bridge



Pin Headers



Mounting Holes



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Title: UE0112-WS28B20 Matrix v8

SKU: UE0112

REV: 1

Last date time: 08/06/2025 04:37 p. m.

SHEET: 3 of 3

File:

Sheet description: Connectors

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